AYAN GHOSH

Agarpara, West Bengal, 700109 | +91 6294268556 | reenaranighosh1974@gmail.com | Portfolio | LinkedIn | GitHub

EDUCATION

Narula Institute of Technology, Agarpara, West Bengal Aug 2024 – June 2028
B.Tech, Electronics & Communications Engineering CGPA: 7.78

Kalyani Public School, Barasat, West Bengal May 2022 – April 2024
Higher Secondary Education Grade: 81.6%

SKILLS

Languages: Python, C++, C, Java, SQL
Electronics & Embedded Systems: Arduino, ESP32, Embedded C
Circuit Design & Simulation: Proteus
Machine Learning & AI: NumPy, Pandas, NLP
Communication & Networking: Digital Electronics, Analog Electronics, Signals & Systems, Communication Systems
Tools & Version Control: Git, GitHub, VS Code, Google Colab
Core Subjects: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks

EXPERIENCE

Data Intern — *Proxie Studio Private Limited* Nov 2025 – Dec 2025

- Conducted data research and lead generation across multiple industries.
- Collected, verified, and organized business data using spreadsheets and Notion.
- Maintained accurate datasets to support sales and business development teams.
- Improved data quality through validation, cleaning, and categorization.

Student Member — *Institution of Electronics and Telecommunication Engineers (IETE)* 2025 – Present

- Active member of the IETE Student Chapter, participating in technical seminars, workshops, and webinars.
- Enhanced knowledge of electronics, communication systems, and emerging technologies.
- Developed teamwork, communication, and problem-solving skills through chapter activities.

PROJECTS

Class AB Power Amplifier Design and Simulation
Tech Stack: Tinkercad Circuits, UA741 Op-Amp, TIP31/TIP32 BJTs

- Designed and simulated a Class AB power amplifier on a ±9 V dual supply, with a UA741 non-inverting driver stage (gain ≈ 5.7).
- Implemented a complementary push-pull output stage with TIP31 (NPN) and TIP32 (PNP) transistors and 1 Ω emitter resistors, driving an 8 Ω load.
- Used a 1N4007 diode bias network to bias the output transistors and reduce crossover distortion.
- Applied a 1 kHz, 0.5 V sine input from a function generator and verified amplification on a virtual oscilloscope.

Spam Message Detection using Machine Learning
Tech Stack: Python, Pandas, NumPy, Scikit-learn, TF-IDF

- Developed a machine learning model to classify SMS messages as Spam or Ham using NLP.
- Performed text preprocessing: cleaning, tokenization, stop-word removal, and stemming; extracted features using TF-IDF Vectorization.
- Trained and evaluated multiple ML models, selecting Logistic Regression for high classification accuracy.
- Built an interactive prediction interface to classify user-entered messages in real time.

CERTIFICATIONS

- Mobile App Development for Android
- Data Internship Certificate
- Introduction to Database Systems (NPTEL)
- IETE Student Membership Certificate